1	Solution: Let *x*, *y*, *z* be the amount needed for Food A, B and C respectively, $0.6x+0.4y+0.3z=110$ $0.3x+0.3y+0.7z=85$ $0.1x+0.3y=45$ Using plysmult, $x=86.25,\quad y=121.25,\quad z=32.5$ Food A =86.25g, Food B = 121.25, Food C= 32.5g Amount of money needed = 86.25($0.60) + 121.25($0.40) + 32.5($0.30)=$110	
2	(i) $\mathrm{D}_f=(-\infty,\infty)$ and $\mathrm{R}_g=(2,\infty)$ Since $\mathrm{R}_g=(2,\infty)\subset \mathrm{D}_f=(-\infty,\infty) \Rightarrow$ function *fg* exists. $fg(x)=f\left[g(x)\right]=\left(\sqrt{x+4}\right)^2-1$ $fg(x)=x+3,\quad x\in\Re^+$ (ii) Since $\mathrm{R}_{fg}=(3,\infty)\subset \mathrm{D}_g=(0,\infty) \Rightarrow$ function *gfg* exists $h(x)=gfg(x)=g\left[fg(x)\right]=\sqrt{(x+3)+4}=\sqrt{x+7},\quad x>0$ (iii) $h(x)=g^{-1}g(x),\quad x>0$ $g\left[fg(x)\right]=g^{-1}g(x)$ $\sqrt{x+7}=x,\quad x\in\Re^+$ $x+7=x^2$ $x^2-x-7=0$ $x=\dfrac{1\pm\sqrt{1-4(1)(-7)}}{2(1)}=\dfrac{1}{2}\left(1\pm\sqrt{29}\right)$ $x=\dfrac{1}{2}\left(1+\sqrt{29}\right)\quad \sin ce\ x\in\Re^+$	
3	$\begin{pmatrix}1+\lambda\\ \lambda\\ 1-2\lambda\end{pmatrix}=\begin{pmatrix}1+7\mu\\ 2+5\mu\\ -1+\mu\end{pmatrix}$ $\lambda-7\mu=0\quad(1)$ $\lambda-5\mu=2\quad(2)$ $2\lambda+\mu=2\quad(3)$ Solving (1) and (2): $\mu=1,\lambda=7$ (3): 2(7)+1=15$\neq 2$ No unique solution and the lines are not parallel. Therefore, they are skew lines.	

	Let $\vec{OM} = \begin{pmatrix} 1+\lambda \\ \lambda \\ 1-2\lambda \end{pmatrix}$ for some $\lambda \in \mathbb{R}$. $\vec{AM} \cdot \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} \lambda \\ \lambda-2 \\ 2-2\lambda \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} = 0$ $\lambda + \lambda - 2 - 4 + 4\lambda = 0 \Rightarrow \lambda = 1$ $\lvert \vec{AM} \rvert = \left\lvert \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \right\rvert = \sqrt{2}$ units **OR** Let B = (1, 0, 1). $\vec{BA} = \begin{pmatrix} 0 \\ 2 \\ -2 \end{pmatrix}$ Shortest distance = $\dfrac{1}{\sqrt{1+1+4}} \left\lvert \begin{pmatrix} 0 \\ 2 \\ -2 \end{pmatrix} \times \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} \right\rvert$ $= \dfrac{1}{\sqrt{6}} \left\lvert \begin{pmatrix} -2 \\ -2 \\ -2 \end{pmatrix} \right\rvert$ $= \sqrt{2}$ units	
4	$y = e^{\sin^{-1}(x)}$ Differentiate wrt to x $\dfrac{dy}{dx} = e^{\sin^{-1}(x)} \dfrac{1}{\sqrt{1-x^2}} = \dfrac{y}{\sqrt{1-x^2}}$ Squaring both sides $\left(1-x^2\right)\left(\dfrac{dy}{dx}\right)^2 = y^2$ Differentiate wrt to x $\left(1-x^2\right)2\left(\dfrac{dy}{dx}\right)\dfrac{d^2y}{dx^2} + \left(\dfrac{dy}{dx}\right)^2(-2x) = 2y\dfrac{dy}{dx}$ $\left(1-x^2\right)\dfrac{d^2y}{dx^2} + \left(\dfrac{dy}{dx}\right)(-x) = y$ Differentiate wrt to x $\left(1-x^2\right)\dfrac{d^3y}{dx^3} + \dfrac{d^2y}{dx^2}(-2x) + \left(\dfrac{d^2y}{dx^2}\right)(-x) - \dfrac{dy}{dx} = \dfrac{dy}{dx}$ $\left(1-x^2\right)\dfrac{d^3y}{dx^3} - 3x\dfrac{d^2y}{dx^2} = 2\dfrac{dy}{dx}$ When x = 0, y = 1, $\dfrac{dy}{dx} = 1$, $\dfrac{d^2y}{dx^2} = 1$, $\dfrac{d^3y}{dx^3} = 2$ $y = 1 + x + \dfrac{x^2}{2} + \dfrac{x^3}{3}$	

$$e^{\sin^{-1}(x)} = 1 + x + \frac{x^2}{2} + \frac{x^3}{3} \dots$$

Using $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3}$

$$\sin^{-1}(x) = \ln\left(1 + x + \frac{x^2}{2} + \frac{x^3}{3}\right)$$

$$= \left(x + \frac{x^2}{2} + \frac{x^3}{3}\right) - \frac{1}{2}\left(x + \frac{x^2}{2} + \frac{x^3}{3}\right)^2 + \frac{1}{3}\left(x + \frac{x^2}{2} + \frac{x^3}{3}\right)^3$$

$$= \left(x + \frac{x^2}{2} + \frac{x^3}{3}\right) - \frac{1}{2}\left(x^2 + x^3 + \cdots\right) + \frac{1}{3}\left(x^3 + \cdots\right)$$

$$= x + \frac{1}{6}x^3$$

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(i) from gc $x \approx 1.70$

(ii) taking limits on both sides

$$x = (10 - 3x)^{\frac{1}{3}}$$

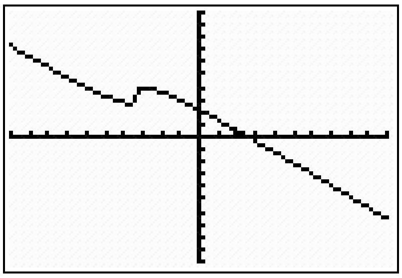
$$x^3 = 10 - 3x$$

$$x^3 + 3x - 10 = 0$$

Thus the limit satisfies the equation in (i). Thus the sequence will converge to α

(iii)

$$x_{n+1} - x_n = (10 - 3x_n)^{\frac{1}{3}} - x_n = f(x_n)$$



From graph

If $x_n < \alpha$, then $f(x_n) > 0$

$\therefore x_{n+1} - x_n > 0 \Rightarrow x_{n+1} > x_n$

If $x_n > \alpha$, then $f(x_n) < 0$

$\therefore x_{n+1} - x_n < \Rightarrow x_{n+1} < x_n$

(iv)

$$y - y_1 = m(x - x_1)$$

$$y - f(x_n) = f'(x_n)(x - x_n)$$

$$-f(x_n) = f'(x_n)(x_{n+1} - x_n)$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{x_n^3 + 3x_n - 10}{3x_n^2 + 3}$$

$$= \frac{2x_n^3 + 10}{3x_n^2 + 3}$$

6(i)

$$\frac{r-1}{r(r+1)}-\frac{r}{(r+1)(r+2)}=\frac{(r-1)(r+2)-r^2}{r(r+1)(r+2)}$$
$$=\frac{r-2}{r(r+1)(r+2)}$$

$$\frac{2}{3\cdot4\cdot5}+\frac{4}{4\cdot5\cdot6}+\frac{6}{5\cdot6\cdot7}+\ldots+\frac{2(N-2)}{N(N+1)(N+2)}=2\sum_{r=3}^{N}\frac{r-2}{r(r+1)(r+2)}$$
$$=2\sum_{r=3}^{N}\left[\frac{r-1}{r(r+1)}-\frac{r}{(r+1)(r+2)}\right]$$
$$=2\left[\frac{2}{3\cdot4}-\frac{3}{4\cdot5}\right.$$
$$+\frac{3}{5\cdot6}-\frac{4}{6\cdot7}$$
$$\vdots$$
$$+\frac{N-2}{(N-1)N}-\frac{N-1}{N(N+1)}$$
$$\left.+\frac{N-1}{N(N+1)}-\frac{N}{(N+1)(N+2)}\right]$$
$$=\frac{1}{3}-\frac{2N}{(N+1)(N+2)}$$

$$\sum_{r=1}^{N-2}\frac{r}{(r+2)(r+3)(r+4)}=\frac{1}{3\cdot4\cdot5}+\frac{2}{4\cdot5\cdot6}+\ldots+\frac{N-2}{N(N+1)(N+2)}$$
$$=\sum_{r=3}^{N}\frac{r-2}{r(r+1)(r+2)}$$
$$=\frac{1}{2}\left[\frac{1}{3}-\frac{2N}{(N+1)(N+2)}\right]$$
$$<\frac{1}{6}\quad\text{since }\frac{2N}{(N+1)(N+2)}>0$$

(ii)

$$u_r-u_{r-1}=-\frac{2(r-2)}{r(r+1)(r+2)}$$
$$\sum_{r=2}^{n}(u_r-u_{r-1})=-2\sum_{r=2}^{n}\frac{r-2}{r(r+1)(r+2)}$$
$$u_n-u_1=-2\sum_{r=3}^{n}\frac{r-2}{r(r+1)(r+2)}$$
$$=-\frac{1}{3}+\frac{2n}{(n+1)(n+2)}$$
$$u_n=\frac{2n}{(n+1)(n+2)}-\frac{1}{6}$$

7

(i) Since C has a vertical asymptote at $x=0$, thus $\lambda+1=0,\Rightarrow\lambda=-1$.

(ii) $y=\dfrac{2x^2+4}{3x}=\dfrac{2}{3}x+\dfrac{4}{3x}$

Oblique asymptote is $y=\dfrac{2}{3}x$

(iii) $y=\dfrac{2x^2+4}{3x}$

$3xy=2x^2+4$

$2x^2-3xy+4=0$

The equation above has no real roots when discriminant < 0.

$\left(-3y\right)^2-4\left(2\right)\left(4\right)<0$

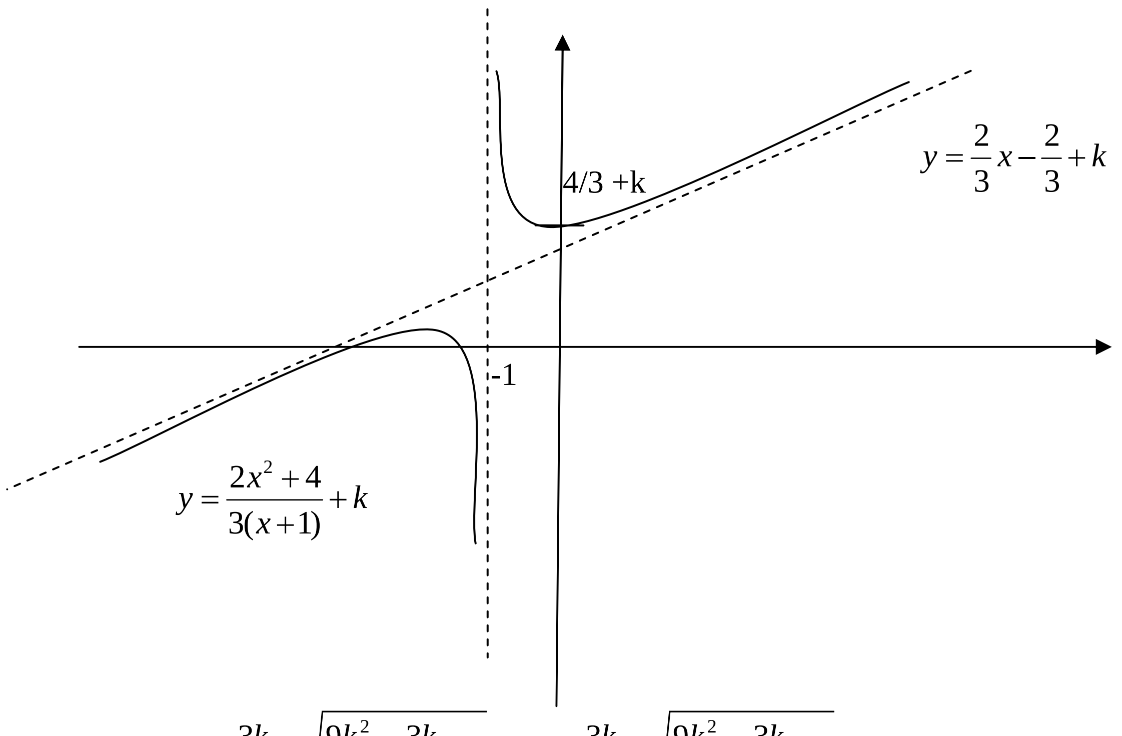
$9y^2-32<0$ M1, for getting b^2-4ac <0 correctly

$(y+\dfrac{\sqrt{32}}{3})(y-\dfrac{\sqrt{32}}{3})<0$

$-\dfrac{4\sqrt{2}}{3}<y<\dfrac{4\sqrt{2}}{3}$

$\therefore a=2$.

(iv) $y=\dfrac{2x^2+4}{3(x+1)}+k=\dfrac{2}{3}x-\dfrac{2}{3}+k+\dfrac{6}{3(x+1)}$



$x-$ intercepts are $-\dfrac{3k}{4}+\sqrt{\dfrac{9k^2}{16}-\dfrac{3k}{2}-2}$ and $-\dfrac{3k}{4}-\sqrt{\dfrac{9k^2}{16}-\dfrac{3k}{2}-2}$

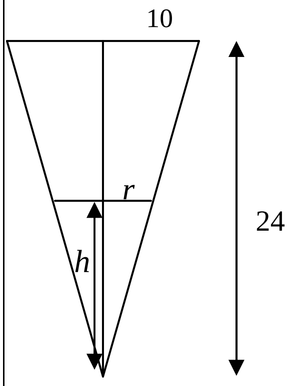
(v) For the graph of $y=\dfrac{2x^2+4}{3(x+1)}+k$ to cut the x-axis at 2 distinct points,

$\sqrt{\dfrac{9k^2}{16}-\dfrac{3k}{2}-2}>0$, =>least integer value of k is 4

8a	$\int x\tan^{-1}\left(2x^2\right)\mathrm{d}x$ $=\frac{1}{2}x^2\tan^{-1}\left(2x^2\right)-\int\frac{2x^3}{1+4x^4}\,\mathrm{d}x$ $=\frac{1}{2}x^2\tan^{-1}\left(2x^2\right)-\frac{1}{8}\int\frac{16x^3}{1+4x^4}\,\mathrm{d}x$ $=\frac{1}{2}x^2\tan^{-1}\left(2x^2\right)-\frac{1}{8}\ln\left(1+4x^4\right)+C$	$u=\tan^{-1}\left(2x^2\right)$ $\quad \frac{\mathrm{d}V}{\mathrm{d}x}=x$ $\frac{du}{dx}=\frac{4x}{1+4x^4}$ $\quad V=\frac{x^2}{2}$	
8b	$3\int_0^m\frac{1}{\pi\sqrt{1-9x^2}}\,\mathrm{d}x=\frac{3}{3\pi}\int_0^m\frac{1}{\sqrt{\left(\frac{1}{3}\right)^2-x^2}}\,\mathrm{d}x$ $=\frac{1}{\pi}\left[\sin^{-1}\left(3x\right)\right]_0^m=\frac{1}{\pi}\sin^{-1}\left(3m\right)$ $\frac{1}{\pi}\sin^{-1}\left(3m\right)=\frac{1}{4}$ $\Rightarrow\sin^{-1}\left(3m\right)=\frac{\pi}{4}\Rightarrow 3m=\frac{\sqrt{2}}{2}$ $\therefore m=\frac{\sqrt{2}}{6}$		
8c	$5k^2-3x^2=2x^2\Rightarrow x=\pm k$ Volume $=2\pi\int_0^k\left(5k^2-3x^2\right)^2-\left(2x^2\right)^2\,\mathrm{d}x$ $=2\pi\int_0^k\left(25k^4-30k^2x^2+9x^4\right)-4x^4\,\mathrm{d}x$ $=2\pi\int_0^k\left(25k^4-30k^2x^2+5x^4\right)\mathrm{d}x$ $=2\pi\left[25k^4x-10k^2x^3+x^5\right]_0^k$ $=2\pi\left(25k^5-10k^5+k^5\right)$ $=32\pi k^5$		
9a	$S_n=S_\infty-S_n$ $2S_n=S_\infty$ $2\frac{a\left(1-r^n\right)}{1-r}=\frac{a}{1-r}$ $2\left(1-r^n\right)=1$ $r^n=\frac{1}{2}$ Given $ar^n=\frac{1}{2}$, hence $\quad a=1$ $ar^{2n}=\left(\frac{1}{2}\right)^2=\frac{1}{4}$		
(b)	$\frac{a+2d}{a+7d}=\frac{a+d}{a+2d}$ $3d^2=-4ad$		

	$d=-\frac{4}{3}a \quad (d\neq 0)$ $r=\dfrac{a-\frac{4}{3}a}{a-\frac{8}{3}a}=\frac{1}{5}$ Sum of even-numbered terms $=\dfrac{\frac{10}{5}}{1-\left(\frac{1}{5}\right)^2}=\frac{25}{12}$ $\ln\lvert u_n u_{n+1}\rvert-\ln\lvert u_{n-1}u_n\rvert=\ln\left\lvert\dfrac{u_n u_{n+1}}{u_{n-1}u_n}\right\rvert$ $=\ln\left\lvert\dfrac{u_{n+1}}{u_{n-1}}\right\rvert$ $=2\ln\left(\frac{1}{5}\right)$ (constant) common difference $=-2\ln 5$	
10i	$w^3=-i$ $=e^{i(-\frac{\pi}{2}+2k\pi)}$ $w=e^{i(\frac{-\pi+4k\pi}{6})}\ ,k=-1,0,1$ $w=e^{i(\frac{-5k\pi}{6})},e^{i(\frac{-\pi}{6})},e^{i(\frac{\pi}{2})}$	
10ii)	$(i-z)^3=iz^3$ $\dfrac{(i-z)^3}{z^3}=i$ $\dfrac{(z-i)^3}{z^3}=-i$ $\dfrac{z-i}{z}=e^{i\theta},\theta=\dfrac{k\pi}{6}$ $z-i=ze^{i\theta}$ $z(1-e^{i\theta})=i$ $z=\dfrac{i}{1-e^{i\theta}}$ $=\dfrac{i}{1-e^{i\frac{k\pi}{6}}}$ (shown) From (i) k = -5, -1, 3	

10iii	$z=\frac{i}{1-e^{i\frac{k\pi}{6}}}$ $=\frac{i}{e^{\frac{ik\pi}{12}}(e^{-\frac{ik\pi}{12}}-e^{\frac{ik\pi}{12}})}$ $=\frac{ie^{-\frac{ik\pi}{12}}}{-2i\sin(\frac{k\pi}{12})}=\frac{i(\cos(\frac{k\pi}{12})-i\sin(\frac{k\pi}{12}))}{-2i\sin(\frac{k\pi}{12})}$ $=\frac{1}{2}+i\frac{1}{2}\cot(\frac{k\pi}{12})$ The roots $\frac{1}{2}-i\frac{1}{2}\cot(\frac{5\pi}{12}),\frac{1}{2}-i\frac{1}{2}\cot(\frac{\pi}{12}),\frac{1}{2}+i\frac{1}{2}\cot(\frac{\pi}{4})$	
11(i)	When x = 0, $2t^2+t=0$ $t(2t+1)=0$ $t=0,t=-\frac{1}{2}$ The points are $(0,0),(0,{}^{11}\!/_{8})$ When y = 0 $t^3-3t=0$ $t(t^2-3)=0$ $t=0,t=\pm\sqrt{3}$ The points are (0,0), ($6\pm\sqrt{3}$,0)	
(ii)	$\frac{dx}{dt}=4t+1$, $\frac{dy}{dt}=3t^2-3$, $\frac{dy}{dx}=\frac{3t^2-3}{4t+1}$ Tangent parallel to the y-axis $4t+1=0 \Rightarrow t=\frac{-1}{4}$ $(\frac{-1}{8},\frac{47}{64})$ Equation of tangent $x=-\frac{1}{8}$	
(iii)	At $t=-\frac{1}{2}$ $x=0,y=\frac{11}{8},\frac{dy}{dx}=\frac{9}{4}$ Equation of tangent $y-\frac{11}{8}=\frac{9}{4}x$ $t^3-3t-\frac{11}{8}=\frac{9}{4}\left(2t^2+t\right)$	

	Solving $t=\frac{11}{2},-\frac{1}{2}$ At $t=\frac{11}{2}$, the point is $\left(66,\frac{1199}{8}\right)$	
(b)	 $\frac{r}{h}=\frac{10}{24}$ $r=\frac{5}{12}h$ $V=\frac{1}{3}\pi r^2 h$ $=\frac{1}{3}\pi\left(\frac{5}{12}h\right)^2 h=\frac{25}{432}\pi h^3$ $\frac{dV}{dt}=\frac{25}{144}\pi(h)^2\frac{dh}{dt}$ When h = 16, $\frac{dV}{dt}=20$, $\frac{dh}{dt}=0.143$ m/s	